Lesson Notes

MODULE 3 | LESSON 4

VOLATILITY AND CORRELATIONS

Reading Time	30 minutes
Prior Knowledge	Volatility, Correlation, Diversification, Bond Prices, Bitcoin
Keywords	Asset Allocation ETFs, Take a hit

In the previous lesson, we outlined the properties, benefits, and weaknesses of exchange-traded funds. We looked at examples of market-wide and sector-specific ETFs. In this lesson, we look at asset allocation ETFs. Then, we illustrate some days where securities took a hit, revealing that correlations may increase when volatility increases, resulting in a loss of diversification when it is most needed.

1. Market Corrections

In the previous two lessons, we discussed the benefits of diversification and saw how ETFs can provide those benefits. One type of ETF we have not yet discussed is asset allocation ETFs. These ETFs own different asset classes. For example, AOR is the iShares Core Growth Allocation ETF with investments in equities and fixed income. Managed by iShares, this fund invests in other iShares ETFs. We started this module thinking about the difference between securities and portfolios. Here is a single security that is essentially a portfolio of *other* portfolios of securities. Such a security is often referred to as a fund-of-funds. This fund tracks a proprietary index and has a mix of equities and fixed incomes.

The benefit is an extra level of diversification. An investor could decide for themselves how to construct a set of ETFs that performs asset allocation. However, the do-it-yourself approach requires insight and experience both to select and to manage these portfolios. The benefit of these asset allocation funds is extra diversification, but at the added expense of an extra layer of management fees. Whether these fees are justified remains a study for the prudent investor.

Moreover, does having multiple asset classes really protect an investor? Let's suppose you held bonds, equities, and Bitcoin at the beginning of 2022. For many asset classes, the financial markets started the year off with quite a bit of turbulence.

1.2 Bonds Take a Hit

Suppose you held fixed income bonds in the United Kingdom, where they are known as gilts. On January 18, 2022, the UK gilt market had high yields. Their yields rose to the highest level since the COVID-19 pandemic began in anticipation of the Central Bank raising interest rates (Ritchie).

Recall that when interest rates go up, bond prices come down. Higher interest rates mean that we discount cash flows more deeply.

1.3 Bitcoin Takes a Hit

January 2022 wasn't a great month for Bitcoin. Sometimes, January is a great month for Bitcoin. In pre-pandemic times, specifically January 2020, Bitcoin had its best close in seven years after increasing 30%. Two years later, the results were very different. From January 18-22, 2022, Bitcoin took a big hit. On January 18, it closed at $42,390. Over the next four days, it dropped to $35,103. Computing the return, we get

= -17.2% return.

Hold on. If you bought on January 22 and held for 15 days, the price on February 6 closed at $42,423. The return from January 22 to February 6 is

= +20.9%.

It took four days to lose 18.9% and another 2 weeks to gain it back! Bitcoin has some serious volatility.

1.4 The Technology Sector Takes a Hit

On February 3, 2022, technology ETFs, like Invesco, SMH, and ARKW, showed a 2.4% decline in a single day. NASDAQ was down 3% that day. Similarly, the Technology Select SPDR, XLK, tumbled. Even though there are numerous stocks in an ETF, if they are concentrated in a sector, they are subject to risks that affect the entire industry.

1.5 Meta Takes a Hit

Also on February 3, 2022, Meta (formerly known as Facebook) fell 26%, wiping out $251 billion in value. This is approximately the annual GDP of Finland, with a population of 5.5 million people. Meta set a record for the largest drop in market capitalization in history.

2. Correlations

Clearly, we know that there are high correlations within the same asset class. In Financial Econometrics, you will look at the correlations of different fixed-income maturities and find that maturities close together are highly correlated. You'll see that there is even a statistical technique, called principal component analysis, that exploits the high correlations in a correlation matrix to provide insight into common sources of variation.

Even within the same asset class, there are high correlations within sectors, especially when the industry as a whole is impacted by a common factor. For example, the pandemic brought the travel industry to a near halt. Stocks from the travel industry suffered big losses at the outbreak of the pandemic in March 2020.

Less clear, however, is that there is correlation across asset classes. Note in the examples above that the drops happened in three different asset classes, spread out over a few weeks. One important observation is that volatility was also high during that time. Indeed, correlations can change when volatility is high.

2.2 Volatility and Correlation

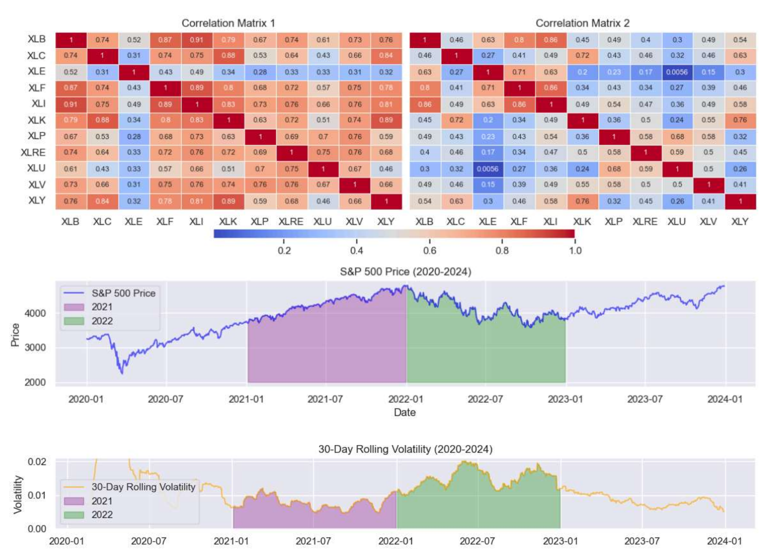
So what is the relationship between volatility and correlation? When discussing volatility, most investors agree that volatility changes over time. There's even an index called the VIX that measures the volatility of members of the SP100. Indeed, excess kurtosis is a reminder that standard deviations are not good representatives of volatility because they don't express volatility well for non-normal distributions.

2.3 Correlations Change

Is there a simple measure for changing correlations? Not exactly. Variances and standard deviations are properties of individual distributions. Correlations only come into play when discussing two distributions. Correlations and variances combine to tell us covariances. Covariances can be on different scales, depending on the sizes of the individual variances (think Bitcoin volatility compared to fixed income volatility). Correlations are scaled: their values always lie within minus 1 to 1. However, correlations can change over time. One way to understand this is to examine the relationship between correlation and volatility. When reading Loretan and English, notice that the relationship between volatility and correlation depends on the level of volatility: when volatility is low, correlations tend to be lower between two series than when those series have high volatility.

This sobering fact affects what we know about diversifying a portfolio. Diversification, the very heart of what low correlation promises, can be yanked away when it is most needed: in highly volatile markets.

Figure 1: Heatmap Correlation of Sectors during Two Distinct Periods



Each correlation matrix is calculated on the S&P500 sectors during two distinct periods: 2021 and 2022, as seen in purple *and green, respectively. Can you identify (without doing calculations) the correlation matrix that corresponds to each period?*

3. Conclusion

Volatility and correlation, the subject of our last two modules, are tightly intertwined and are often the key criteria a financial engineer will study when trading and hedging. We have seen ways to measure them, but this is just the beginning. In the next module, we turn to another issue that causes returns to be volatile: leverage. We'll also add a new asset class to our toolkit: options.

References

- Loretan, Mico, and William B. English. "Special Features: Evaluating Changes in Correlation during Periods of High Market Volatility." *BIS Quarterly Review*, June 2000, pp. 29–36. https://www.bis.org/publ/r_qt0006e.pdf.
- Ritchie, Greg. "Rate-Hike Mania Hits UK as 10-year Yield Climbs to 2019 High." *Bloomberg*, 18 Jan. 2022, https://www.bloomberg.com/news/articles/2022-01-18/rate-hike-mania-hits-u-k-as-10-year-yield-climbs-to-2019-high